MILITARY SPECIFICATION

SEALING COMPOUND, NON-CARCINOGEN CURING POLYURETHANE (POLYETHER-BASED), ELECTRIC CONNECTORS AND ELECTRIC SYSTEMS

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a two-component, elevated temperature curing, reversion resistant polyether polyurethane compound for sealing and embedding electrical and electronic components in Military Weapons Systems. Component materials and cured compound shall not contain any known carcinogens.

1.2 Classification. The sealing compound shall be of the following types and classes as specified (see 6.2).

Type I - Two component
Type II - Premixed, degassed and frozen
Class 1 - Hardness point range of 65±5
Class 2 - Hardness point range of 80±5
Class 3 - Hardness point range of 90±5

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8030

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SPECIFICATIONS

Federal

QQ-A-250/4	Aluminum Alloy, 2024, Plate and Sheet
QQ-A-250/13	Aluminum Alloy Alclad 7075, Plate and Sheet
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Fiberboard
PPP-C-96	Can, Metal, 28 Gage and Lighter

Military

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-R-6855	Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes
MIL-S-8660	Silicone Compound
MIL-S-18729	Steel Plate, Sheet and Strip Alloy 4130, Aircraft Quality
MIL-W-22759/9	Wire, Electric, Fluorocarbon-Insulated, Extruded TFE, Silver-Coated Copper Conductor, 1000-Volt
MIL-L-23699	Lubricating Oil, Aircraft Turbine Engines, Synthetic Base

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MS3100	Connector, Receptacle, Electric, Wall Mounting
MS3103	Connector, Receptacle, Electric for Potting
MS3106	Connector, Plug, Electric, Straight
MS25183	Connector, Plug, Electric, for Potting

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM D 149	Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies
ASTM D 150	A-C Loss Characteristics and Dielectric Constant (Permittivity) of Solid Electrical Insulating Materials
ASTM D 257	D-C Resistance or Conductance of Insulating Materials
ASTM D 395	Compression Set of Vulcanized Rubber
ASTM D 412	Tension Testing of Vulcanized Rubber
ASTM D 495	High-Voltage, Low-Current, Arc Resistance of Solid Electrical Insulating Materials
ASTM D 518	Surface Cracking Resistance of Stretched Rubber Compounds
ASTM D 624	Tear Resistance of Vulcanized Rubber
ASTM D 792	Specific Gravity and Density of Plastics by Displacement
ASTM D 1149	Accelerated Ozone Cracking of Vulcanized Rubber
ASTM D 2240	Indentation Hardness of Rubber and Plastics by Means of a Durometer

(Application for copies of ASTM Standards should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Classification of requirements. The requirements for the sealing compound are classified herein as follows:

Requirement	Paragraph
Material	3.2
Primer	3.2.1
Appearance	3.2.2
Toxicity	3.3
Color	3.4
Properties before cure	3.5.1
Properties after cure	3.5.2
Storage life	3.6

3.2 Material. The sealing compound shall be a two-component system consisting of a prepolymer of the polyether polyurethane type and a curing agent before mixing. The prepolymer shall cure by the addition of the curing agent and shall not depend on solvents or moisture for curing.

3.2.1 Primer. The use of a primer to improve the adhesion of the sealing compound is permitted. The primer shall be recommended by the manufacturer and shall be included in the tests of this specification. The primer shall be identified on the applicable Qualified Products List (QPL). The use and application of the acceptable primer shall be in strict accordance with the manufacturer's instructions.

3.2.2 Appearance. Each component, and mixed compounds shall have a uniform homogeneous texture and shall be free of coarse particles, foreign matter and lumps. Storage of the materials shall cause no separation or crystallization of ingredients which cannot be readily re-dissolved on heating to approximately 49°C (120°F).

3.3 Toxicity. Each component and mixed uncured compound, or cured compound shall have no adverse effect on the health of personnel engaged in the mixing and application of the systems under conditions of adequate ventilation and while exercising caution to avoid prolonged contact with the skin and while observing Occupational Safety and Health Administration (OSHA) guidelines (see 6.4).

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3.4 Color. Unless otherwise specified in the contract or order, the compound shall be furnished in the color manufactured.

3.5 Characteristics and performance

3.5.1 Properties before curing

3.5.1.1 Nonvolatile content. The nonvolatile content of the mixed compound shall be not less than 99 percent when tested as specified in 4.5.1.1.

3.5.1.2 Application life. The requirements for application life shall include the determination of the initial viscosity of mixed type I compound, initial flow rate of type II compound when brought to room temperature, and the application time.

3.5.1.2.1 Initial viscosity of mixed type I compound. When tested as specified in 4.5.1.2.1, the required initial viscosity range for each Class of type I compound shall be 8,000 to 12,000 centipoises.

3.5.1.2.2 Initial flow rate (type II). When tested as specified in 4.5.1.2.2, the initial flow rate of type II compound for each class shall be 200 to 900 grams per minute.

3.5.1.2.3 Application time. The application time of the mixed compound shall be not less than 90 minutes (minimum specified time to reach 250,000 centipoises) when tested as specified in 4.5.1.3.

3.5.2 Properties after cure.

3.5.2.1 Physical properties. The physical properties of the cured compound shall conform to the requirements of table I.

3.5.2.2 Electrical properties. The electrical properties of the cured sealing compound shall conform to the requirements of table II.

3.6 Storage life

3.6.1 Type I. The manufacturer shall certify that the type I compound after storage in the original unopened container at standard conditions (4.4.1) for 180 days when mixed and cured as specified in 4.4.2 and 4.4.5 respectively, shall conform to the requirements of tables I and II.

3.6.2 Type II. The manufacturer shall certify that the type II compound after storage -34.4±1.1°C (-30±2°F) for 60 days when cured as specified in 4.4.5, shall conform to the requirements of tables I and II.

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TABLE I

Physical properties of cured compounds.

Physical property	Requirements Types I and II			Test paragraph
	Class			
	1	2	3	
Hardness, Type A Durometer 1/	65±5	80±5	90±5	4.5.1.4
Tensile Strength, PSI, minimum	1600	3300	4000	4.5.1.5
Ultimate elongation, percent, min.	400	400	400	4.5.1.5
Shrinkage percent, maximum	5.0	5.0	5.0	4.5.1.6
Specific gravity	1.0-1.2	1.0-1.2	1.0-1.2	4.5.1.7
Tear strength, ppi, min.	130	200	340	4.5.1.8
Compression set, percent max.	35	50	65	4.5.1.9
Adhesion, lbs/inch of width, min.				
To steel	30	40	50	4.5.1.10
To aluminum	30	40	50	4.5.1.10
Moisture absorption, percent max.	3.0	3.0	3.0	4.5.1.11
Low temperature flexibility at -53.9°C (-65°F) 2/	Pass	Pass	Pass	4.5.1.12
Fungus resistance	Non-nutrient	Non-nutrient	Non-nutrient	4.5.1.13
Ozone resistance, 28 days exposure 3/	Pass	Pass	Pass	4.5.1.14
Oil resistance:				
Change in hardness, percent max.	40	40	40	4.5.1.15
Change in weight, percent max.	40	30	30	4.5.1.15
Heat aging:				
Change in hardness, percent max.	15	15	15	4.5.1.16
Tensile strength after heat aging, PSI, min.	1500	3200	3500	4.5.1.16
Elongation, Percent, min.	300	300	300	4.5.1.16
Hydrolytic stability:				
Change in hardness points, max.	15	25	25	4.5.1.17
Tensile strength after exposure, PSI, min.	1400	2300	3000	4.5.1.17
Elongation, percent min.	300	300	300	4.5.1.17
Corrosion 4/	Pass	Pass	Pass	4.5.1.18

1/ Test specimens shall be cured as specified in 4.4.5.
2/ There shall be no evidence of cracking, blistering or loss of adhesion at -53.9°±1.1°C (-65°±2°F).
3/ There shall be no evidence of cracking as a result of ozone exposure when viewed under 7X magnification.
4/ There shall be no evidence of pitting or corrosion of the copper. Discoloration of the copper will be permitted.

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inch (12.7 mm) deep. Five aluminum panels, each 4 by

TABLE II

Electrical properties. 1/

Tests	Requirements Type I and II			Test paragraph
	Class I	Class II	Class III	
Resistance to arc, sec., min.	50	50	50	4.5.2.1
Dielectric strength, volts/mil, min.	250	250	250	4.5.2.2
Dielectric constant:				
1 kilohertz at 25°C (77°F), max.	10	10	10	4.5.2.3
1 megahertz at 25°C (77°F), max.	8	8	8	
Power factor:				4.5.2.3
1 kilohertz at 25°C (77°F), max.	0.06	0.06	0.06	
1 megahertz at 25°C (77°F), max.	0.10	0.10	0.10	
Volume resistivity, Ohms-CMS:				4.5.2.4
at 25°C (77°F), min.	4×10^{11}	6×10^{11}	1×10^{12}	
at 121°C (250°F), max.	2.5×10^{9}	5×10^{9}	5×10^{9}	
Surface resistivity, Ohms:				4.5.2.4
at 25°C (77°F), min.	2.4×10^{12}	2.4×10^{12}	1×10^{13}	
at 121°C (250°F), min.	2.4×10^{9}	2.4×10^{9}	5×10^{9}	
High potential resistance	No breakdown	No breakdown	No breakdown	4.5.2.5
Insulation resistance, Megohm:				4.5.2.6
Melamine insert				
Initial at 25°C (77°F)	2.0×10^{5}	3.0×10^{5}	3.5×10^{5}	
at 121°C (250°F)	400	700	750	
Neoprene insert				
Initial at 25°C (77°F)	2.0×10^{5}	3.0×10^{5}	3.5×10^{5}	
After hydrolytic stability				
Insulation resistance, megohms, min.	1000	1000	1000	4.5.2.7
High potential	No breakdown	No breakdown	No breakdown	4.5.2.7
Flame resistance overload	Shall not ignite	Shall not ignite	Shall not ignite	4.5.2.8

1/ Based on average compound properties without special processing.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.2 Classification of inspections. The examination and testing of the sealing compound shall be classified as quality conformance (see 4.3).

4.3 Quality conformance

4.3.1 Lot formation. Unless otherwise specified, a lot shall consist of all the material of the same type and class, manufactured at one time from one batch (see 6.3), forming part of one contract or order and submitted for acceptance at the same time and place.

4.3.2 Report of tests. The manufacturer shall submit test reports to the Government representative for each lot showing the results including numerical values when applicable of all the requirements in 3.5.1, 3.5.2, table I and table II. In addition, the manufacturer shall certify to the storage requirements of 3.3 and 3.6. Reports of numerical tests as satisfactory are not acceptable.

4.3.3 Sampling and inspection procedures.

4.3.3.1 Sampling for tests. Samples shall be randomly selected from each lot of sealing compound offered for inspection. Sufficient material shall be selected to complete the testing of all requirements specified in 4.3.2. The material selected shall be visually examined to 3.2.2 and 3.4. Failure of the sealing compound to pass any requirement shall be cause to reject the lot represented by the material.

4.3.3.2 Packaging. A quantity of shipping containers prepared for delivery just prior to closure shall be selected in accordance with Inspection Level I of MIL-STD-105. The lot size for purposes of this inspection shall be the number of shipping containers. The samples shall be examined for packaging, packing, marking, leakage, and workmanship. Any package having one or more defects shall be rejected. If the number of defective packages exceeds the Acceptable Quantity Level (AQL) of 4.0 percent defective in accordance with MIL-STD-105, the lot represented by the sample shall be rejected.

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4.3.3.3 Container fill. The sample size shall be determined in accordance with Inspection Level S-3 of MIL-STD-105 except that the sample size derived from MIL-STD-105 shall be rounded off to the next highest increment of five. At no time shall the sample size be less than five, unless the lot consists of less than five containers in which all containers shall be used to determine the fill. The sample unit shall be one filled container. The lot shall be rejected if the average content per container is less than that specified.

4.4 Test conditions.

4.4.1 Standard conditions. Unless otherwise specified herein, all mixing, and testing shall be performed at 25±1.1°C (77±2°F) and a relative humidity of 50±5 percent.

4.4.2 Mixing of type I compounds. The base compound and its curing agent, both in the original unopened containers, together with the required spatulas, beakers, and other mixing equipment shall be held at standard conditions (4.4.1) for a minimum of 6 hours. The base compound and curing agent shall then be thoroughly mixed in the proportions recommended by the manufacturer. Proper care shall be taken to avoid incorporation of air by excessive stirring. The mixed material shall be degassed under vacuum.

4.4.3 Thawing of type II compounds. The premixed, frozen cartridges of type II compound shall be thawed in a 54.4°C ±2.7°C (130° ±5°F) water bath for 30±3 minutes and then placed at 25°±1.1°C (77°±2°F) for 10± 1 minutes prior to testing.

4.4.4 Molding and potting. Molds for disk specimens and special shapes as indicated in individual test descriptions shall be of polyethylene, polytetrafluoroethylene (PTFE) coated metal, or coated with any other material having a nonadhesive surface which will not transfer to the test specimen. The mold shall be constructed to facilitate removal of the specimen. The mold shall be thoroughly cleaned with a suitable solvent and dried prior to the application of the primer and mold release agent and compound. The mixed compound shall be poured or extruded slowly into the mold. The mold shall then be covered with a flat, smooth plate and a pressure not to exceed 10 pounds per square inch (psi) applied.

4.4.5 Curing. Specimens prepared as specified in 4.4.2 through 4.4.4 shall be cured for 18 hours at 82.2°±1.1°C (180°±2°F). Cure time at standard conditions at least 24 hours prior to testing unless otherwise specified herein.

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4.4.6 Electric connector assemblies. Four connector assemblies equipped with melamine inserts and four assemblies equipped with neoprene inserts, conforming to the following shall be used:

MS3103 Connector, receptacle
MS25183 Connector, plug

Type MS3100 A22-19P Connector, receptacle
Type MS3106 A22-19S Connector, plug

Two foot lengths of polytetrafluoroethylene (PTFE) insulated wire conforming to MIL-W-22759/9 shall be chemically treated to provide a bondable surface, then soldered to each pin of the connector. Insulation resistance determinations are required before potting the wired assemblies (see 4.5.2.5). The mixed polyurethane shall be poured or extruded slowly into the connector until a slight excess is obtained. During all potting operations conformance to 3.5.1.2.3 shall be noted. The assemblies shall be cured for 18 hours at 82.2°±1.1°C (180°±2°F). The cured assemblies shall be kept at standard conditions for 24 hours. Testing shall commence following completion of this conditioning period. All connectors shall have their contact materials identified by means of insulation sleeving or other markers designed for attachment about the wire bundle. The cured potted connector assemblies shall be tested as specified in 4.5.2.5.

4.4.7 Finished specimens. The surfaces of molded and potted polyurethane specimens shall be clean, smooth, free from holes and bubbles, and all corners and edges shall be well defined.

4.5 Test methods

4.5.1 Physical property tests

4.5.1.1 Nonvolatile content. Five to ten grams of the mixed compound shall be transferred as soon after mixing as possible to a dish about 8 centimeters (cm) in diameter. A tight fitting cover shall immediately be placed over the dish and the weight determined to the nearest milligram. The cover shall then be removed and the compound heated for 24 hours at 82.2°±5.5°C (180°±10°F). It shall then be cooled in a closed desiccator, removed from the desiccator, and the weight determined to the nearest milligram. The percent nonvolatile content shall be calculated as follows:

$$\text{Percent Nonvolatile Content} = \frac{\text{Final weight}}{\text{Initial weight}} \times 100$$

4.5.1.2 Initial viscosity and flow rate

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4.5.1.2.1 Initial viscosity (type I). A 250 gram sample of freshly mixed type I compound shall be used to determine viscosity. The sample shall be mixed for 3 minutes and its viscosity determined with a Brookfield viscometer model RVF (or equivalent) using a No. 4 spindle and speed of 10 RPM. Readings shall be recorded when the pointer on the viscometer dial reaches a steady position after a minimum of three revolutions. Retain the sample for determining the application time (4.5.1.3.1).

4.5.1.2.2 Initial flow rate (type II). The cartridge shall be thawed as specified in 4.4.3. The nozzle of the cartridge shall have a 0.125±0.005 inch opening and the application gun set for 85±5 psi back pressure. Materials shall be extruded initially to fill the nozzle and to remove any trapped air. Immediately after the thawing period, extrude the material for ten seconds in a tared container. Determine the flow rate in grams per minute by extruding the material for ten seconds and converting the weight to grams per minute.

4.5.1.3 Application time

4.5.1.3.1 Type I. Determine the viscosity of the sample of 4.5.1.2.1, 15 minutes after completion of mixing, with a Brookfield viscometer model RVF (or equivalent) using a No. 6 spindle and speed of 4 RPM. Supplemental readings shall be made periodically until a reading of 250,000 centipoises is attained, which shall be considered as the end of the application time. The application time shall commence at the end of the mixing period. The spindle shall not be drawn from the material during the test. This sample shall be used to prepare hardness specimens (see 4.5.1.4).

4.5.1.3.2 Type II. Type II compound shall be thawed as specified in 4.4.3. Determine its viscosity with a Brookfield viscometer model RVF (or equivalent) using a No. 6 spindle and speed of 4 PRM by extruding a sufficient amount of material and mixing the sample by hand for 15 seconds. The test procedure specified in 4.5.1.3.1 shall be followed until a reading of 250,000 centipoises is attained, which shall be considered as the end of the application time.

4.5.1.4 Hardness. Eight specimens having 2 by 2 inch minimum face dimensions and a minimum thickness of 0.250±0.003 inch shall be prepared and cured as specified in 4.4.5. The hardness shall be determined with a Type A hardness durometer in accordance with ASTM D2240 using a three second application time. Two specimens each shall be used for specific gravity (4.5.1.7), heat aging (4.5.1.16), oil resistance (4.5.1.15) and hardness after hydrolytic exposure (4.5.1.17).

4.5.1.5 Tensile strength and ultimate elongation. Twelve specimens shall be cut from a sheet of the compound 0.075 ±0.005 inches thick, molded as specified in 4.4.4. The test specimens shall be cut

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using Die "C" and tested in accordance with ASTM D412. Three specimens shall be saved for determining the tensile strength and elongation after heat aging (4.5.1.16), and three specimens for determining tensile strength and elongation after hydrolytic stability (4.5.1.17).

4.5.1.6 Shrinkage. A cubical mold, approximately 1.0 inch on each side and having an open top shall be constructed in accordance with 4.4.4. Its volume at 25°±1.1°C (77°±2°F) and R. H. of 50±5 percent shall be determined. A compound specimen shall be prepared in the mold and cured in accordance with 4.4.5. After curing, the specimen shall be placed in a circulating air oven at 82.2°±2.7°C (180°±5°F) for 96 hours. It shall then be removed, cooled, and its volume at 25°±1.1°C (77°±2°F) determined by the water displacement method. The percent shrinkage shall be calculated as follows:

$$\text{Percent shrinkage} = \frac{V_1 - V_2}{V_1} \times 100$$

Where: V_1 = Volume of mold
V_2 = Final volume of compound

4.5.1.7 Specific gravity. Two test specimens from 4.5.1.4 shall be used for determining the specific gravity. Tests shall be in accordance with ASTM D792.

4.5.1.8 Tear strength. Three tear strength specimens shall be cut from a sheet of the compound 0.075±0.005 inches thick, molded as specified in 4.4.4. The test specimens shall be cut using Die "C" and tested in accordance with ASTM D624.

4.5.1.9 Compression set. Three specimens 0.500±0.005 inches thick and 1.129±0.001 inch in diameter shall be molded in accordance with 4.4.4. The compression set shall be determined in accordance with Method B, compression set under constant deflection of ASTM D395. The test shall be conducted at 70°±1°C (158°±2°F) for 22 hours.

4.5.1.10 Adhesion

4.5.1.10.1 Adhesion to steel and aluminum panels

4.5.1.10.1.1 Preparation of specimens. When required, the surface of the metal panels listed in table III shall be vapor degreased, sand blasted, and solvent wiped with methyl-ethyl-ketone. To the cleaned steel and aluminum surfaces, apply a coat of primer (see 3.2.1) when required. Application, film thickness, drying time and required condi-

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tions shall be specified by the manufacturer. A sufficient quantity of mixed sealing compound shall be applied to the panels to form a coating having a thickness of 0.125 inch ±0.030 inch. A 3 inch by 6 inch area of a 3 inch by 12 inch strip of 60 mesh corrosion resistant metal screen shall be intimately coated with the mixed compound and placed on the coated area of each panel leaving a loose end 6 inches long. The panel assemblies shall be cured in accordance with 4.4.4. One panel assembly for each metal shall be prepared

TABLE III. Panels for adhesion test.

Composition	Specification	Size (Inches)	Number of specimens cut from each panel assembly
Steel, Alloy 4130	MIL-S-18729	3 X 6 X 1/16	2
Aluminum Alloy 2024	QQ-A-250/4	3 X 6 X 1/16	2

4.5.1.10.1.2 Immersion of specimens. Immediately after completion of curing, the test specimens shall be immersed vertically for 7 days ±4hours in a closed glass vessel containing a three percent aqueous sodium chloride solution with 4 inches of the panel exposed to the test fluid and the remainder of the panel exposed to the air vapor mixture. The temperature during the test shall be maintained at 70°±1.1°C (158°± 2°F). At the expiration of the test period the fluid shall be cooled to 25°±1.1°C (77°±2°F). The panels shall then be removed and the adhesion determined in accordance with 4.5.1.10.1.3, within 10 minutes after removal from the fluid.

4.5.1.10.1.3 Testing of specimens. Upon completion of 4.5.1.10. 1.2, the specimens shall be tested as follows: Two 1-inch wide strips shall be cut lengthwise through the screen and compound to the panel surface and extended the full length of the loose end of the screen. The edges of the panel shall not be used as one edge of the test strip. The specimens shall be tested individually in an autographic testing machine with a capacity such that the tension at failure is not more than 85 percent and not less than 15 percent of the full scale load. The rate of separation of the jaws shall be 12 inches per minute and shall be uniform at all times. Specimens shall be mounted in the machine so that the loose end of the 1-inch wide screen strip will be folded 180 degrees as it is pulled from the panel. Each strip shall be pulled individually. The adhesion in pounds shall be automatically recorded on a chart as a continuous curve. The adhesion value shall be calculated by averaging the maximum and minimum forces required to separate the compound from the panel. If cohesive failures occur, the adhesion value shall be reported as greater than the measured value.

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4.5.1.11 Moisture absorption. A specimen 1.0 inch by 1.0 inch by 1.0 inch shall be molded in accordance with 4.4.4. The specimen shall be weighed and immersed for 24 hours in a distilled water bath held at 93.3°±2.7°C (200°±5°F). The specimen shall be removed, blotted dry and weighed. The percent of water absorption shall be calculated as follows:

$$\text{Percent } H_2O \text{ Absorption} = \frac{\text{Final Weight - Initial Weight}}{\text{Initial Weight}} \times 100$$

4.5.1.12 Low temperature flexibility. Three test panels of aluminum alloy alclad conforming to QQ-A-250/13, measuring 1 inch by 6 inches by 0.32 inch, shall be coated on one side with the primer specified in 3.2.1 to a dry film thickness of 0.4 to 1.0 mil. A quantity of molding compound shall be mixed and applied to the primed surfaces in a manner which will produce cured coatings with a thickness of 0.050 to 0.060 inch, leaving 1 inch at each end of the panels uncoated. The test specimens shall then be inserted in a flexibility fixture as shown in Figures 1 and 2 so that the uncoated side will contact the contour block and the weight will contact only the uncoated end of the panel. The flexibility fixture and test specimens shall be subjected to a conditioning temperature of -53.9°±1.1°C (-65°±2°F) for 6 hours. After the specified conditioning, the specimens shall be bent around the curved portion of the flexibility test jig by releasing the fastening hook. The test specimens shall be removed and examined for failure due to blistering, cracking, or loss of adhesion.

4.5.1.13 Fungus resistance. Three specimens 4 by 4 by 0.125 inch shall be molded and cured as specified in 4.4.4 and 4.4.5 respectively. Tests shall be conducted in accordance with MIL-E-5272.

4.5.1.14 Ozone resistance. Test specimens shall be molded and cured as specified in 4.4.4 and 4.4.5 to meet table IV requirements. Test equipment shall be in accordance with ASTM D1149. Test procedures shall be in accordance with table IV.

TABLE IV. Ozone resistance specimens and procedure.

Test Variable	Requirement
Test specimen 1/	ASTM D518, Procedure B
Ozone concentration	50±2 pphm (volume)
Exposure period	28 days
Exposure temperature	40°±1°C (104°±2°F)
Magnification	7X

1/ Two specimens required.

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4.5.1.15 Oil resistance. Two specimens from 4.5.1.4 shall be weighed to the nearest milligram and immersed in lubricating oil conforming to MIL-L-23699 for a period of 24 hours ±10 minutes at 93.3°±1.1°C (200°±2°F). The oil bath containing the specimens shall then be cooled to standard conditions (4.4.1). Determine the hardness and change in weight of the specimens after oil immersion for conformance to table I.

4.5.1.16 Heat aging. Test specimens prepared as specified in 4.5.1.4 and 4.5.1.5 shall be exposed in an air circulating oven for 168 hours at 100°±2°C (212°±5°F). The specimens shall then be removed and cooled for one hour at standard conditions (4.4.1). Determine the change in hardness (4.5.1.4), tensile strength (4.5.1.5), and ultimate elongation (4.5.1.5) after heat-aging for conformance to table I.

4.5.1.17 Hydrolytic stability

4.5.1.17.1 Specimens. The specimens prepared in 4.5.1.5 shall be used for the determination of the tensile strength after hydrolytic exposure. Hardness shall be determined on the specimens of 4.5.1.4 before and after hydrolytic exposure at the same location on the specimens using a 3 second application time.

4.5.1.17.2 Procedure. Test specimens shall be placed vertically in a suitable holder, on a tray in a suitable glass desiccator (250 mm.). The desiccator shall contain a glycerine (22 percent by weight) in water solution in the bottom which will produce a minimum relative humidity (RH) of 95 percent at the test temperature. The desiccator, containing the specimens, shall then be closed and inserted into an air circulating oven maintained at 93.3°±1°C (200°±2°F) for a period of 28 days. At the end of the exposure period, the desiccator shall be removed from the oven and cooled to standard conditions (4.4.1) for 16 to 24 hours. The averaged results for the hardness, tensile strength, and ultimate elongation shall be recorded and checked for conformance to table I.

4.5.1.18 Corrosion. A polished pure copper strip approximately 1 inch wide by 3 inches long of a convenient thickness shall be covered on one side with the sealing compound and allowed to remain for 48 hours. At the end of this period, the compound shall be removed from the copper strip and the strip examined for evidence of corrosion. Primer, when specified, by the manufacturer shall be coated on a separate copper strip as noted above and allowed to remain for 48 hours. The primer shall be removed from the strip and the copper strip examined for evidence of corrosion. Pitting or corrosion of the copper shall be cause for rejection. Discoloration of the copper shall not be considered objectionable.

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4.5.2 Electrical tests

4.5.2.1 Arc resistance. Arc resistance shall be determined in accordance with ASTM Method D495. Test specimens shall be prepared as specified in table V and 4.4.4. The specimens shall be smooth and free from dust.

4.5.2.2 Dielectric strength. Test specimens shall be prepared as specified in table V and 4.4.4. Dielectric strength determinations shall be made in accordance with ASTM Method D149. Electrodes shall be brass cylinders, 2 inches in diameter and 1 inch thick with edges rounded to 1/4 inch radius, and shall be applied to the opposite faces of the specimens. The tests shall be made under oil at a frequency not exceeding 100 cycles per second (CPS). The voltage shall be increased uniformly at the rate of 500 volts per second.

4.5.2.3 Dielectric constant and power factor. Test specimens shall be prepared and cured as specified in table V and 4.4.4. Tests shall be conducted in accordance with ASTM Method D150. Lead or tin foil electrodes shall be used and applied to the specimen with a thin film of silicone conforming to MIL-S-8660 or equivalent. The electrodes shall consist of two lead or tin foil disks, one of the same diameter as the specimen and the other 2 inches in diameter, centrally located on the opposite face of the specimen. The test current shall be introduced to the foil through two brass discs. The upper disc shall be 2 inches in diameter by 1 inch in thickness, and the lower disc shall be 4 inches in diameter. The specimens shall be tested at frequencies of 1 Kilohertz and 1 Megahertz at a temperature of 25°±1°C (77°±2°F). Calculation shall include corrections for edge and ground capacitance.

TABLE V. Disc specimens.

Property	Test para. no.	Disc size (inch)		Number of discs
		Diameter	Thickness	
Arc Resistance	4.5.2.1	4	0.125	3
Dielectric strength	4.5.2.2	4	0.125	3
Dielectric Constant & Power Factor	4.5.2.3	4	0.125	3
Volume & Surface Resistivity	4.5.2.4	4	0.125	3

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4.5.2.4 Volume and surface resistivity. Disc specimens shall be prepared as specified in table V and 4.4.4. Resistivity tests shall be conducted in accordance with ASTM Method D257, using a General Radio Type 544B bridge or equivalent instrument with a test voltage of 500 volts. Readings shall be made 1 minute after application of current. Lead or tin foil electrodes shall be a disc, 2 inches in diameter, centrally located on one face of the specimen. The guard electrode shall be a concentric ring 2.281 inches inside diameter and with an outside diameter equal to that of the specimen. The unguarded electrode shall be a foil disc 4 inches in diameter applied to the opposite side of the specimen. The test current shall be introduced to the guarded electrode, the guard electrode, and the unguarded electrode by means of a brass disc 2 inches in diameter by 1 inch in thickness, a brass ring 2.312 inches inside diameter by 4 inches outside diameter by 0.125 inch thick, and a brass disc 4 inches in diameter, respectively. Calculations necessary for volume and surface resistivity shall be made in accordance with ASTM D257 for effective area and effective perimeter. Tests shall be conducted at standard conditions and at 121°±1°C (250°±2°F), after a conditioning period of 30 minutes at that temperature.

4.5.2.5 High potential resistance. Connector assemblies shall be tested for high potential resistance before and after hydrolytic stability (see 4.5.2.7). A potential of 2000 volts root mean square (RMS), 60 cycles, shall be applied between all contacts and shell for a period of one minute. The test voltage shall be applied at the rate of approximately 500 volts per second until the specified voltage is reached.

4.5.2.6 Insulation resistance. Insulation resistance shall be determined on the wired, unpotted assemblies. All connector assemblies potted as specified in 4.4.6, regardless of cure procedure, shall undergo insulation resistance testing at 25°±1°C (77°±2°F) (4.5.2.6.1), and after hydrolytic stability (4.5.2.7).

4.5.2.6.1 Procedure. Insulation resistance of each assembly shall be determined by a megohm bridge such as General Radio Type 544B or equivalent using a potential of approximately 500 volts. Electrification time shall not exceed 2 minutes. Tests shall be conducted at 25°±1°C (77°±2°F) and at 121°±1°C (250°±2°F) after the assemblies have been conditioned for 30 minutes at that temperature. Two sets of readings shall be taken. The first set shall be between the shell and contacts A, E, and G; the second set shall be made between contacts L to P and M to N. The average value of each set of readings shall be reported independently. No single value within each set shall be less than the minimum specified value of table II. When a single value deviates from the average by more than 20 percent, additional determinations shall be made for each set as follows:

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First set readings between the shell and contacts B, C, D, H, J and K.

Second set readings between contacts B to C, D to E, G to H, and J to K.

4.5.2.7 Conditioning for hydrolytic stability, electrical. The mated connector assemblies as specified in 4.5.2.6 shall be tested for insulation resistance and high potential resistance after hydrolytic stability conditioning. The assemblies shall be placed in a humidity test chamber of the type described in 4.5.1.17.2. The ends of the wire leads shall be sealed by wrapping with a glass filament-polyester insulation tape, or equivalent. The assemblies shall be exposed to a relative humidity (RH) of 95 percent at 93°±1°C (200°±2°F) for 28 days ±4 hours. After exposure, the entire test chamber shall be removed from the heat source and cooled to room temperature for 16 to 24 hours before testing for high potential resistance and insulation resistance as specified in 4.5.2.5 and 4.5.2.6. Aluminum foil shall be wrapped around the exposed compound when the assembly is not in the humidity chamber.

4.5.2.8 Flame resistance overload. A cylindrical test specimen 1-1/2 inches long by 1/2 inch diameter shall be prepared. Centered within and along the lengthwise axis of the specimen shall be a 12 inch length of insulated wire (size 16) conforming to MIL-W-22759/9. The lengthwise axis of the specimen shall be centered on the wire. The test specimen shall then be held taut in a horizontal position and 55 amperes of direct current applied for a minimum of 2-1/2 minutes. The specimen shall meet the requirements of table II.

4.5.3 Storage life

4.5.3.1 Type I. A minimum of one quart of base compound and the appropriate amount of curing agent, both in original, unopened containers, shall be stored at standard conditions (4.4.1) for 180 days. The compound shall be evaluated for conformance to 3.6.1.

4.5.3.2 Type II. Premixed frozen cartridges shall be stored for 60 days at -34.4°±1°C (-30°±2°F). After 60 days, the cartridges shall be thawed (4.4.3) and the material checked for conformance to 3.6.2.

5. PACKAGING

5.1 Preservation - packaging. Packaging shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A

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5.1.1.1 Type I compound.

5.1.1.1.1 Unit protection. The base compound and curing agent shall be packaged in their own containers. The curing agent and base compound containers shall be attached in such a way as to allow easy separation for mixing purposes but will prevent accidental separation. The ratio of the quantity contained in the attached curing agent container shall be the same as the recommended mixing ratio of the base compound and curing agent.

5.1.1.1.2 Unit package. The base compound shall be furnished in 1/2 pint, 1 pint, 1 quart or 1 gallon friction top cans, conforming to Type V, Class 1 of PPP-C-96.

5.1.1.2 Type II compound. Type II compound shall be supplied premixed, degassed and quick frozen in six and twelve ounce plastic cartridges complete with cap or plug, as manufactured by Pyles Industries, Inc. and Semco Research Co., or equal as specified in the contract or order. (see 6.2).

5.1.2 Commercial. Type I base compound with curing agent, and Type II compound shall be packaged in accordance with the manufacturer's commercial practice.

5.1.3 Instructions for use. The supplier of the sealing compound shall furnish with each unit package, printed label instructions containing the following:

(a) Storage life, uncatalyzed, indicating the optimum storage temperature and conditions of storage to obtain maximum storage life.

(b) The proportions of compounds and curing agents of Type I materials shall be on a weight basis and the methods for mixing them shall be clearly stated.

(c) Recommended thawing instructions for type II materials.

5.2 Packing. Packing shall be level A or Commercial as specified (6.2).

5.2.1 Level A

5.2.1.1 Type I compound. The base compound and curing agent packaged in accordance with 5.1.1.1.2, shall be packed in an overseas exterior type containers conforming to PPP-B-585, PPP-B-601, PPP-B-621, or PPP-B-636. Weight of the contents shall conform to the weight limitation of the applicable box specification and shall not exceed a total

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weight of 200 pounds. Containers shall be closed and strapped in accordance with Appendix to PPP-C-96.

5.2.1.2 Type II compound. Type II unit packages of 5.1.1.2, frozen for shipment, shall be packed in a manner that will prevent thawing or other damage during transit. Shipping containers shall contain the same number of the type of unit packages, be of uniform size, and shall be designed to insure that damage is prevented during handling and shipping.

5.2.2 Commercial. Type I and II materials packaged in accordance with 5.1.2 shall be packed to afford protection against damage during direct shipment from the source of supply to the first receiving activity for immediate use. Containers shall comply to the Uniform Freight Classification Rules or other regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special markings required by the contract or order, unit packages, intermediate packages, and exterior shipping containers shall be marked in accordance with MIL-STD-129. Marking shall include specification number, type, and class; name of manufacturer; date of manufacture and expected shelf life expiration date (month and year for Type I material and month, day and year for Type III material) and recommended storage temperature.

6. NOTES

6.1 Intended use. The sealing compounds covered by this specification are intended for potting or encasing electrical connectors and electronic components. Used within its operating temperature range (-54° to +93°C continuous, +121°C intermittent) (-65° to +200°F continuous, +250°F intermittent), the cured sealing compound acts as a deterrent to fatigue, corrosion, contamination, and aids in reducing arc-over between pins of an electrical connector.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type and Class of sealing compound required (see 1.2).

(c) Quantity desired.

(d) Size of container.

(e) Level of packaging and packing required (see 5.1 and 5.2).

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(f) Any special marking required.

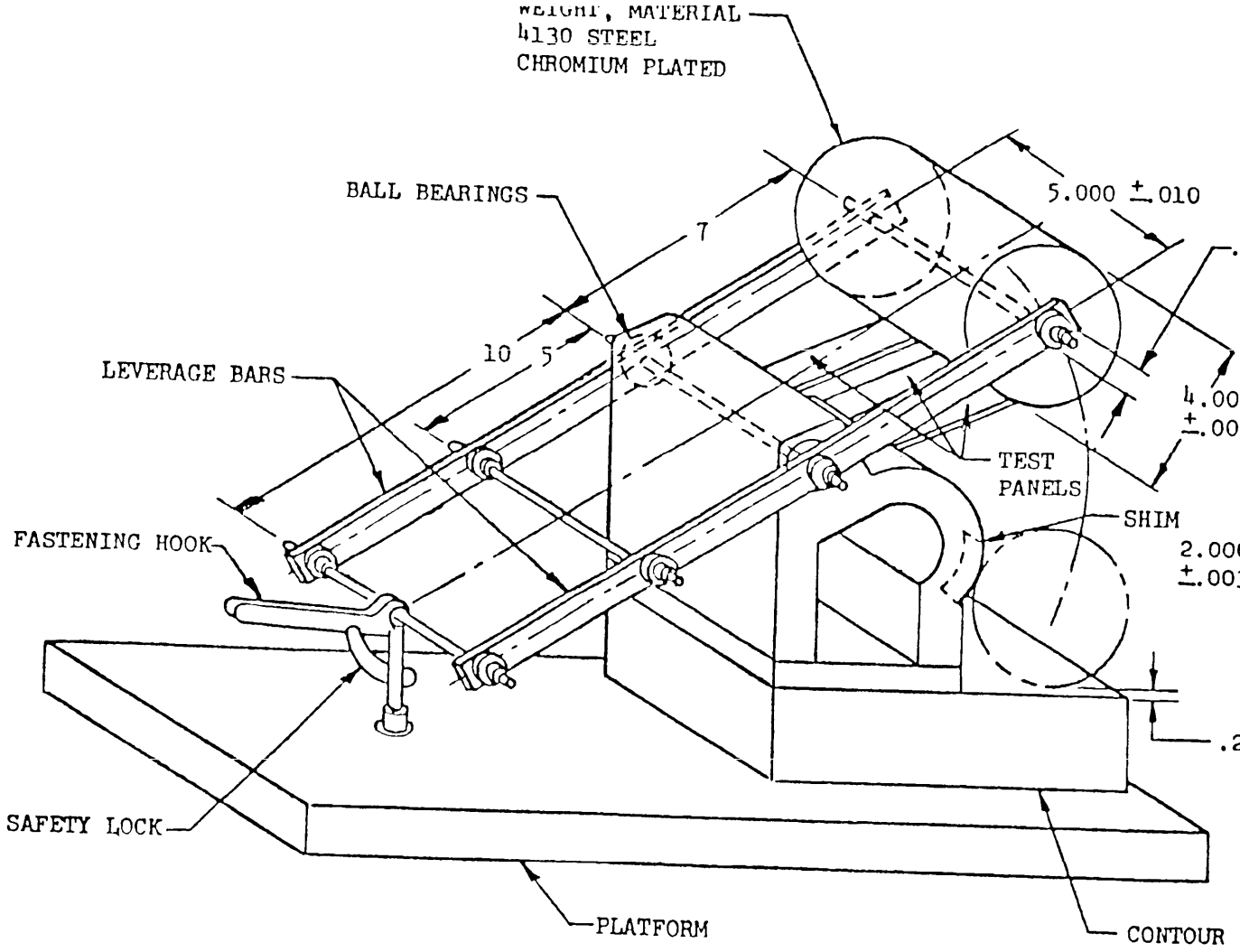
(g) Additional requirements, if applicable (e.g., color).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Toxicity. Questions pertaining to the requirements of 3.3 should be referred, by the procuring activity, to the Navy Environmental Health Center, 3333 Vine Street, Cincinnati, Ohio 45520, who will act as an advisor to the procuring activity.

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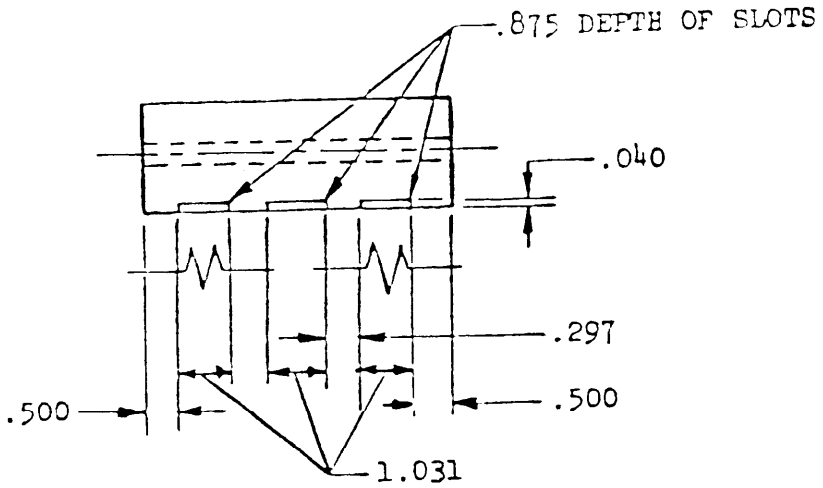
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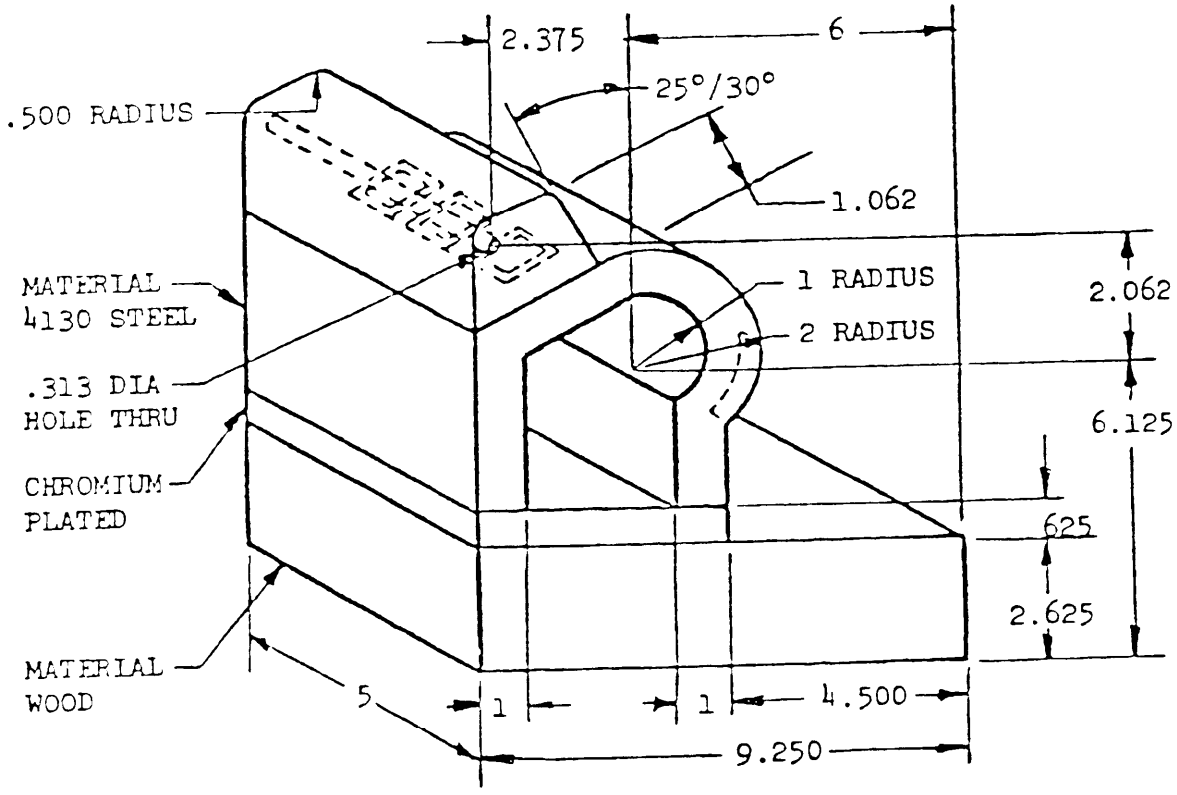
Dimensions are in inches.

Figure 1. Low Temperature Test Fixture

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VIEW SHOWING SLOTS FOR PANELS



Dimensions are in inches.

Figure 2. Contour Block

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